

250th Anniversary of US Independence



Source: PIB

The Prime Minister of India extended greetings to the President of the US, and the people of the US on the historic occasion of the **250 th anniversary of the country's independence (4 th July 2026)**.

- **US Declaration of Independence:** The US celebrates Independence Day on **4 th July** commemorating the adoption of the **Declaration of Independence (1776)**, a landmark document that proclaimed the independence of the 13 American colonies from British rule and laid the foundation for modern democracy, constitutionalism, and human rights.
 - The US Declaration of Independence, drafted primarily by Thomas Jefferson in June 1776 as the principal author of the Committee of Five, proclaimed that **all individuals are created equal and are endowed with unalienable rights, including Life, Liberty, and the Pursuit of Happiness**.
 - It formally challenged the "divine right of kings" and established that governments derive their just powers from the "**consent of the governed**."
- **Impact:** US Declaration of Independence served as the **intellectual foundation for modern human rights** , influencing the **Universal Declaration of Human Rights (UDHR), 1948** .
- **Influence on India:** The US Constitution's Bill of Rights, a direct descendant of the Declaration's push for individual liberties, served as the primary model for the **Fundamental Rights enshrined in Part III of the Indian Constitution** , particularly **Articles 14 and 21** .

Read more: [Navigating India-US Ties](#)

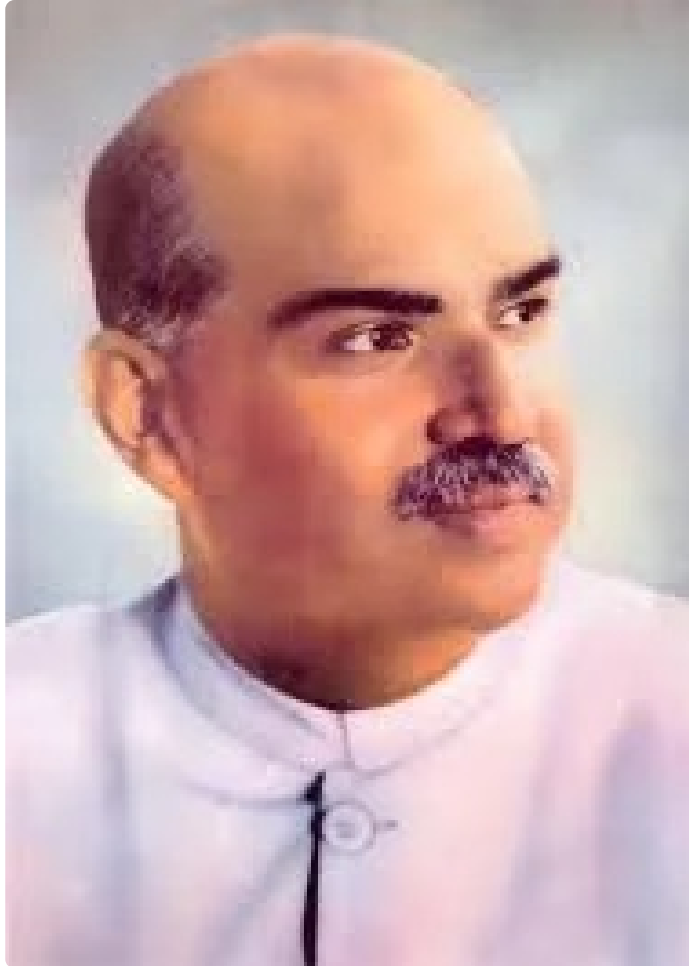
Birth Anniversary of Dr. Syama Prasad Mookerjee



Source: PIB

The Prime Minister paid tribute to **Dr. Syama Prasad Mookerjee** on his birth anniversary (**6 th July**) highlighting his legacy in national unity, industrial policy, and education, and his relevance in India's modern development.

- **Birth and Early Life** : Born in **Calcutta** (now Kolkata). Dr. Mookerjee is the son of **Sir Ashutosh Mookerjee** , a renowned educationist and Vice-Chancellor of **Calcutta University** .
- **Academic Excellence** : Dr. Mookerjee represented **Calcutta University** at the **Conference of the Universities of the British Empire** in England.
 - Dr. Mookerjee became the **youngest Vice-Chancellor** of **Calcutta University** at the age of 33 in **1934** .
 - He started "**Bang Wani**", a Bengali journal, in 1922 and The **Nationalist** in the 1940s.
- **Political Career** : In the 1920s, Dr. Mookerjee joined the Indian National Congress but resigned later over ideological differences with the leadership. He later joined the Hindu Mahasabha and played a key role in forming a Progressive Coalition government in Bengal in 1937, where he served as the **Finance Minister**.
 - Became the **acting President** of the **Hindu Mahasabha** in 1940 and advocated for **complete independence** for India.
 - He founded the **All India Bharatiya Jan Sangh** in **1951** , which later evolved into the **Bharatiya Janata Party**.
- **Post-Independence Role** : Dr. Mookerjee served as **Minister for Industry and Supply** in the **Interim Government** after independence.
 - Instrumental in establishing key institutions like the **Chittaranjan Locomotive Factory** , **Sindri Fertilizer Corporation** , and **Hindustan Fertilizer Corporation** .
- **Ideological Stance**: He advocated nationalism, Hindu cultural identity, and a united India. He opposed Article 370, asserting that one nation **cannot have two constitutions, heads, or flags**.
 - **Opposed the** linguistic division of India, advocating for unity based on **administrative efficiency, security, and economic prosperity** .
 - He was arrested during a protest against J&K's special status and died under mysterious circumstances in 1953.
- **Legacy** : He is known as "**The Lion of Parliament**" for his sharp debates on national issues.



Read more: [Dr. Shyama Prasad Mukherjee](#)

PM Pays Tribute to Swami Vivekananda



Source: PIB

The Prime Minister paid tributes to **Swami Vivekananda on his Nirvana Day**. He highlighted Swami Vivekananda's contribution to bringing global recognition to India's cultural heritage, spiritual traditions and national consciousness.

- **About:** Swami Vivekananda, born as Narendra **Nath Datta** on **12 th January 1863**, was a monk and the chief disciple of **Ramakrishna Paramhansa** .
 - In **1893** , upon the request of **Maharaja Ajit Singh** of the Khetri State, he took the name '**Vivekananda**', changing from '**Sachidananda**' that he used before.
- **Enlightenment:** In **1892** , Swami Vivekananda was said to have swum to a **rock** (later named **Vivekananda Rock Memorial**) in the Indian Ocean from the shores of **Kanyakumari for**

meditation.

- He spent **three days and nights there** , resulting in his **enlightenment**.

- **Contributions:**

- **Philosophical:** He introduced the world to the Indian philosophies of **Vedanta** and **Yoga** .
 - He preached ‘ **neo-Vedanta** ’ , an interpretation of **Hinduism through a Western lens** , and believed in combining **spirituality with material progress**.
- **Spiritual:** Vivekananda's message on human values draws from the **Upanishads** , the **Gita** , and the examples of **Buddha** and **Jesus**, emphasizing **self-realization, compassion, and selfless service** .
 - He advocated the doctrine of **service** . Serving **jiva** (living beings) is considered **worship of Shiva**.
 - He gave the **four pathways** of attaining **moksha (liberation)** from the worldly pleasure and attachment in his books- **Raja-yoga, Karma-yoga, Jnana-yoga and Bhakti-yoga**.
- **Revivalism:** He laid emphasis on **education** for the **regeneration** of our motherland. He advocated a **man-making character-building education**.
- **Nationalism:** His nationalism is based on **Humanism and Universalism** , the two cardinal features of Indian spiritual culture.
 - His nationalism is based on concern for the masses, freedom, equality, and **Karma Yoga** — a path to **political and spiritual freedom** through selfless service.
- **Associated Organisations:** He founded the **Ramakrishna Mission in 1897** to propagate the ideals of **service, education, and spiritual upliftment**.
 - In **1899** , he established the **Belur Math**, which became his permanent abode.
- **International Address:** He addressed the **Parliament of Religions held in Chicago in 1893** , at which he represented **Hinduism**.
 - In July, **1896**, he addressed a conference of the **London Hindu Association** in London.
- **Death:** He passed away on **4th July 1902**, while meditating at **Belur Math** . His death anniversary is observed as **Nirvana Day**.



Read more: [Swami Vivekananda](#)

Creation of a Lab-Made "Cell"



Source: TH

Researchers in the US have achieved a major milestone in **synthetic biology** by developing SpudCell, a lab-made synthetic cell built entirely from non-living chemical components that can grow, replicate its DNA, divide, and undergo evolutionary selection.

- **Constructed from Non-Living Chemicals:** Unlike traditional genetic engineering that modifies existing living organisms, this artificial cell was built entirely from scratch using lifeless chemicals, starting with a basic artificial fat bubble (**liposome**) acting as the cell membrane.
- **Artificial Protein Factory:** The synthetic cell houses a specialized, cell-free chemical mixture (**PURE system**) that successfully translates a minimal, custom-made genetic code (**90,000 base-pair DNA genome**) into working proteins.
- **Nutrient Absorption (Feeding):** The cell's DNA instructs it to create a molecular "hook" (**alpha-hemolysin protein**) on its surface, allowing it to capture and fuse with smaller nutrient bubbles to absorb the lipids and materials it needs to expand.
- **DNA Replication:** As the cell grows in size, it utilizes a specific copying enzyme to accurately duplicate its entire genetic blueprint.
- **Innovative Cell Division:** Instead of relying on a complex internal structural network (**cytoskeleton**) like natural cells, scientists engineered the cell to divide into two daughter cells using simple physical pressure and mechanical stress created by proteins crowding its outer surface.
 - SpudCell shows life-like behaviour, but it is not a fully living or autonomous organism. It needs external supply of food and ribosomes and cannot sustain itself like a natural cell.
- **Darwinian Evolution:** The experiment successfully proved that artificial systems can undergo **natural selection** ; when researchers introduced a genetic mutation that allowed some cells to feed faster, these efficient variants reproduced quicker and eventually outcompeted the original population.
- **Significance:** This is a landmark achievement in **Synthetic Biology** . It proves that basic life processes can be chemically engineered from non-living matter.
 - This technology holds immense future applications for **precision medicine (like programmable drug delivery systems)**, **sustainable industrial biomanufacturing**, and **creating artificial organisms for environmental cleanup** .

Read more: [Synthetic Biology](#)

